

# ECON 0100 | Vignette B2 | Supply

## | Solutions

*Answers and work are shown in the red blocks.*

### Q1 | Consumer Surplus

Members of the wizarding world have preferences for pumpkin pasties according to the following demand curve:

$$P_d = 12 - \frac{1}{2}Q_d$$

Prices are in galleons and quantity is in pasties.

a) Plot and find the consumer surplus at a price of 10 galleons. \_\_\_\_\_

4 galleons.

**Plot.** Vertical intercept,  $Q_d = 0$ :  $P_d = 12$ . Horizontal intercept,  $P_d = 0$ :  $0 = 12 - \frac{1}{2}Q_d \rightarrow Q_d = 24$ .

Price to quantity, then the triangle below demand and above the price:

$$10 = 12 - \frac{1}{2}Q_d \rightarrow \frac{1}{2}Q_d = 2 \rightarrow Q_d = 4$$

$$CS = h \cdot b \cdot \frac{1}{2} = (12 - 10) \cdot 4 \cdot \frac{1}{2} = 4$$

b) Plot and find the consumer surplus at a price of 5 galleons. \_\_\_\_\_

49 galleons.

The same steps at the lower price:

$$5 = 12 - \frac{1}{2}Q_d \rightarrow \frac{1}{2}Q_d = 7 \rightarrow Q_d = 14$$

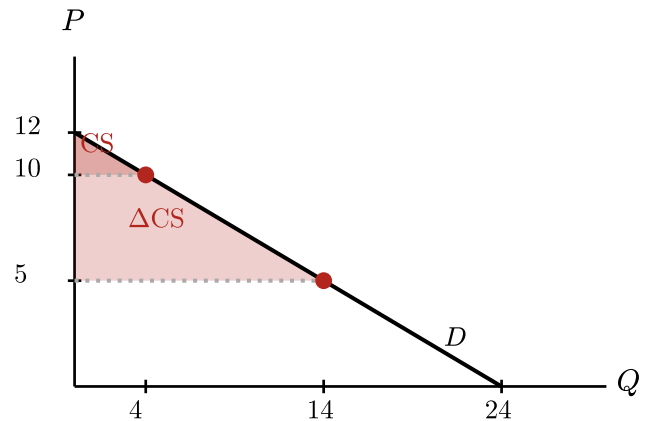
$$CS = h \cdot b \cdot \frac{1}{2} = (12 - 5) \cdot 14 \cdot \frac{1}{2} = 49$$

c) How much did consumer surplus change from (a) to (b)? \_\_\_\_\_

Up by 45 galleons.

The change is the new area minus the old, the strip between the two price lines:

$$\Delta CS = CS - CS = 49 - 4 = 45$$



## Q2 | Quantity Supplied

Pumpkin pasties are produced by many sellers according to the following supply curve:

$$P = 2 + \frac{1}{2}Q_s$$

Plot the supply curve below.

a) Find and plot the quantity supplied at a price of 10 galleons. \_\_\_\_\_

16 pasties.

**Plot.** Vertical intercept,  $Q_s = 0$ :  $P = 2$ . Below 2 galleons quantity supplied is zero, so the curve starts there and rises by 1 galleon for every 2 pasties.

$$10 = 2 + \frac{1}{2}Q_s \rightarrow 8 = \frac{1}{2}Q_s \rightarrow Q_s = 16$$

b) Find and plot the quantity supplied at a price of 5 galleons. \_\_\_\_\_

6 pasties.

$$5 = 2 + \frac{1}{2}Q_s \rightarrow 3 = \frac{1}{2}Q_s \rightarrow Q_s = 6$$

c) How much did quantity supplied change from (a) to (b)? \_\_\_\_\_

Down by 10 pasties.

$\Delta Q_s = 6 - 16 = -10$ : quantity supplied fell by 10 pasties as the price fell, the law of supply.

d) What is marginal cost at 8 pasties? \_\_\_\_\_

6 galleons.

Quantity to price. The height of the supply curve at 8 pasties is the marginal cost of the eighth pastry:  $P = 2 + \frac{1}{2} \cdot 8 = 6$  galleons.

### Q3 | Producer Surplus

Plot and calculate the producer surplus at both 5 galleons and 10 galleons.

a) PS at 5 galleons: \_\_\_\_\_

9 galleons.

Producer surplus is the triangle above supply and below the price, out to the quantity sold from Q2. Its height is the price minus the 2-galleon intercept and its base is the quantity.

$$P = 5 : \text{ PS} = h \cdot b \cdot \frac{1}{2} = (5 - 2) \cdot 6 \cdot \frac{1}{2} = 9$$

b) PS at 10 galleons: \_\_\_\_\_

64 galleons.

$$P = 10 : \text{ PS} = h \cdot b \cdot \frac{1}{2} = (10 - 2) \cdot 16 \cdot \frac{1}{2} = 64$$

c) How much did producer surplus change from (a) to (b)? \_\_\_\_\_

Up by 55 galleons.

$\Delta PS = PS - PS = 64 - 9 = 55$ : producer surplus rose by 55 galleons as the price rose from 5 to 10. Revenue at 10 galleons is the rectangle  $10 \cdot 16 = 160$  the 64 above the curve is surplus and the 96 below it covers marginal costs.

